

Bayesian Network-Based Estimation and Comparison of Player Skill Over Time

CSE 516 — Probabilistic Graphical Models | Milestone 1 (M1): Proposal & Scoping | Group 3

Category: Theoretical AI

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1. Problem Statement

1.1 Selected Problem and Mathematical Question

We study how the choice of inference method affects Bayesian estimates of time-varying player skill, using TrueSkill Through Time (TTT) as the core probabilistic model and methodological focus. The project is centered on one PGM problem: posterior inference over latent player skills that evolve over time from observed match outcomes.

1.2 Motivation and Significance

Elo and the original TrueSkill update a player's rating using information available up to the current match. This forward-only filtering can leave early estimates uncertain and can make historical comparisons difficult because later results cannot revise earlier skill estimates. TrueSkill Through Time addresses this limitation by connecting a player's historical activities in one temporal Bayesian model, allowing information to propagate both backward and forward in time. The central practical question is whether the additional computational cost of smoothing is justified by better estimates and predictions.

1.3 Mathematical Question

How much does allowing information to flow in both directions through a temporal Bayesian network improve skill estimates compared with a forward-only filtering approach, and what computational cost is introduced by the inference methods required to obtain those estimates?

1.4 Assumptions

- A player's performance in a single match is approximately Gaussian-distributed around their hidden skill.
- A player's skill changes gradually over time and is modelled using a Gaussian random-walk dynamics assumption.
- The graph structure is determined by the observed match schedule and temporal ordering; the project focuses on parameter learning rather than structure learning.
- For team matches, team performance is formed from the participating players according to the chosen TTT formulation.

1.5 Scope

We focus on one algorithm family, TrueSkill Through Time, and investigate inference within its probabilistic model rather than surveying unrelated PGM algorithms. The main methodological comparison is between tractable/exact inference on controlled cases, loopy belief propagation, and expectation propagation for the non-Gaussian outcome component. Representation and parameter learning are specified explicitly so that

the investigation covers the full Representation–Inference–Learning (R/I/L) framework required for a PGM project.

2. Representation

2.1 Graphical Model

Player skill is represented using a Dynamic Bayesian Network (DBN). Each player has a latent skill state at each relevant time step. A skill state at time t is connected to the same player's skill at time $t-1$, representing temporal dependence and gradual skill evolution. Within a match, a player's skill generates a performance variable; player performances combine into team performance when necessary; the difference between competing team performances determines the observed match result.

2.2 Variables and Dependencies

The principal latent variable is player skill. Intermediate latent variables are player performance and, for multi-player teams, team performance. The observed variable is the match outcome: win, loss, or draw. Temporal edges encode skill evolution, while within-match edges encode the causal path from skill to performance to outcome. The corresponding Bayesian network is converted into an equivalent factor graph because message-passing inference is naturally expressed using variable and factor nodes.

2.3 Conditional Independencies

The model assumes first-order temporal dependence: a player's current skill depends directly on the previous skill state and the skill-evolution noise rather than directly on every earlier state. Match outcomes depend on the latent skills and performance variables of the participating players or teams. This conditional-independence structure produces the temporal chains and local factors used by message passing.

2.4 Representation Question

The representation question for the project is whether this temporal Bayesian/factor-graph construction captures the dependencies needed for historical skill estimation while remaining computationally manageable as the number of matches and players increases.

3. Inference

3.1 Target of Inference

The goal is to compute the posterior distribution of each player's skill at each point in time, conditioned on observed match outcomes both before and after that point. Unlike forward filtering, TTT can use later observations to revise earlier skill estimates; this is the smoothing property that motivates the project.

3.2 Planned Inference Methods

- Exact sum-product inference on tree-structured Gaussian subgraphs and controlled two-player examples, where Gaussian messages can be computed in closed form.
- Loopy belief propagation (BP) for factor graphs containing cycles, with iterative message updates until a predefined convergence criterion is met.
- Expectation propagation (EP) for the non-Gaussian outcome factor created by win/lose/draw observations, using a Gaussian approximation obtained through moment matching.

3.3 Why Multiple Inference Methods?

The TTT model contains Gaussian temporal and performance components, but the comparison/outcome observation introduces a non-Gaussian factor. In addition, realistic match histories can create cycles in the factor graph. The project therefore separates exact inference on controlled cases from approximate inference on realistic graphs. This lets us study the trade-off among approximation error, convergence, predictive quality, and computational cost.

3.4 Inference Research Question

Under what graph structures and data conditions does bidirectional message passing provide a meaningful improvement over forward-only inference, and when does the additional computation or approximation required by smoothing outweigh its benefit?

4. Learning

4.1 Learning Type

The project performs parameter learning, not structure learning. The graph structure is obtained directly from the match schedule and temporal ordering, while model parameters are estimated or tuned from observed match outcomes.

4.2 Parameters

Parameters to be learned or tuned include the skill-evolution variance, performance/noise variance, initial skill uncertainty, and any draw-related parameter required by the selected TTT formulation. These parameters control how rapidly skill is allowed to change and how uncertain the model remains about latent skill and observed performance.

4.3 Learning Procedure

We plan to estimate the parameters by maximizing model evidence or predictive likelihood on the observed outcomes. Hyperparameter sensitivity will also be examined so that conclusions are not dependent on one arbitrary setting. No learned parameter values or numerical learning results are claimed at M1; these are planned for later milestones.

4.4 Learning Question

How sensitive are posterior skill estimates and predictive performance to the learned dynamics and observation-variance parameters, and does evidence-based tuning improve generalization on held-out outcomes?

5. Baseline

5.1 Reference Baselines

- Original TrueSkill (Herbrich, Minka and Graepel, 2006): the primary Bayesian forward-only filtering baseline. It shares the Gaussian performance model but does not perform the historical smoothing central to TTT.
- Elo: a classical non-Bayesian rating baseline that provides a simple point-estimate reference.
- Constant-skill model: a lower reference point with no temporal learning.

5.2 Purpose of the Baselines

The comparison is deliberately ordered as Baselines → TTT model/inference → recent SOTA/related reference. The original TrueSkill baseline isolates the value of bidirectional temporal inference because it uses a closely related Bayesian skill model while remaining forward-only. Elo and constant skill provide simpler reference points.

6. SOTA and Related Work Position

6.1 SOTA Selection Rationale

For M1, we selected Bayesian Online Natural Gradient (BONG) (Jones, Chang, and Murphy, NeurIPS 2024) as a recent state-of-the-art (SOTA) methodological reference for the Bayesian inference component of our project. Although BONG is not a player-skill rating model, it addresses a closely related methodological problem: performing efficient Bayesian inference for evolving latent quantities under computational constraints. This is relevant to our project because the central research question concerns the trade-off between the quality of posterior skill estimates and the computational cost of the inference procedure, particularly when moving from forward-only filtering to bidirectional message passing and smoothing. BONG therefore provides a contemporary reference point for understanding how modern Bayesian inference methods address accuracy–computation trade-offs. At M1, our SOTA positioning is intentionally scoped around this methodological connection rather than claiming that BONG is a direct replacement or application-level competitor to TTT. As the project progresses and our understanding of the experimental setting becomes more concrete, we may refine the SOTA positioning by adding or substituting a more directly domain-specific recent work, if this provides a stronger comparison with our final research question.

6.2 Recent SOTA Reference for the Inference Component: BONG

BONG is used as the recent methodological SOTA reference for the inference component. Kevin Murphy is affiliated with Google DeepMind; the other authors are affiliated with the University of Colorado Boulder and MIT. BONG is not presented as a direct replacement for TTT or as a player-skill model.

6.2.1 Paper, Authors and Venue

Matt Jones, Peter Chang, and Kevin Murphy, “Bayesian Online Natural Gradient (BONG),” Advances in Neural Information Processing Systems (NeurIPS 37), 2024, Main Conference Track.

6.2.2 Problem Addressed

BONG studies efficient sequential Bayesian inference: updating beliefs over hidden model parameters one observation at a time when exact posterior inference is difficult or unavailable in closed form.

6.2.3 Representation

BONG represents uncertainty using a variational Gaussian distribution, including restricted Gaussian forms such as diagonal-plus-low-rank precision structures. In non-stationary settings, a dynamics model can connect the parameter at one step to the next. This is conceptually related to the skill-over-time chain in TTT, but BONG is formulated for general model parameters rather than player skill.

6.2.4 Learning and Inference

Standard online variational Bayes includes an expected log-likelihood term together with a KL term that keeps the new posterior close to the previous posterior. BONG removes the KL term and instead takes a single natural-gradient step on the expected log-likelihood, initialized from the previous step's prior

prediction. The paper shows that this is exact in conjugate settings and provides strong performance for non-conjugate online inference under controlled computational budgets.

6.2.5 Evaluation and Findings

BONG is evaluated against other online variational Bayes approaches, primarily for online/streaming learning of neural-network parameters. The reported results show that BONG can outperform competing online methods when computational cost is controlled.

6.2.6 Limitations

BONG is designed for general streaming parameter estimation rather than the discrete win/lose/draw outcome structure and explicit multi-team factor graph used in this project. Its single natural-gradient update is also an approximation, and approximation quality can be affected by highly non-Gaussian posteriors.

6.2.7 Relationship to Our Project

The methodological overlap is sequential Bayesian belief tracking under computational constraints. TTT uses an explicit factor graph with message passing and, importantly, smooths by allowing information to propagate backward as well as forward. BONG performs a single natural-gradient update in the forward direction, trading the benefit of smoothing for computational efficiency. This gives the project a clear final comparison axis: message-passing-based smoothing versus single-step online Bayesian update. We will refine this positioning after obtaining our own experimental results; no BONG experimental result is claimed at M1.

6.3 Core TTT Base Paper: 2025

Landfried and Mocsos, “TrueSkill Through Time: Reliable Initial Skill Estimates and Historical Comparability with Julia, Python, and R,” *Journal of Statistical Software*, 2025, is the practical and replication-oriented base paper for this project.

6.3.1 Paper, Authors and Venue

Gustavo Landfried and Esteban Mocsos, *Journal of Statistical Software*, Volume 112, Issue 6, 2025.

6.3.2 Problem Addressed

The paper addresses the limitations of forward-only skill rating by using all historical activities in a single Bayesian network. The objective is to obtain more reliable initial skill estimates, lower uncertainty, and historical comparability by allowing information to flow across time in both directions.

6.3.3 Representation

The paper represents skill as a latent variable evolving over time, with performance and match-outcome variables connecting hidden skill to observed results. The temporal Bayesian network can be expressed as a factor graph suitable for message passing.

6.3.4 Learning

The model contains parameters governing skill dynamics, performance noise, and initial uncertainty. The paper provides a practical implementation framework in Julia, Python, and R and uses real historical tennis data to demonstrate the approach. Our project will study parameter tuning/evidence-based learning as part of the later experimental pipeline rather than claiming learned values at M1.

6.3.5 Inference

TTT uses information from the full temporal history and applies iterative message passing/approximate inference so that past and future observations can influence skill estimates. The paper discusses Gaussian approximations and factor-graph message passing to make inference computationally practical.

6.3.6 Evaluation and Findings

The paper demonstrates the effect of smoothing using controlled examples and an ATP tennis dataset. In small temporal examples, later results can revise earlier estimates, reducing the bias associated with forward-only filtering. The ATP study contains approximately 447,000 matches from 1915–2020 involving more than 19,000 players and is used to demonstrate practical scalability and historical skill estimation.

6.3.7 Limitations and Project Gap

TTT provides the core model and practical implementation, but our project is not simply a replication exercise. We will explicitly compare inference strategies, validate exact inference on controlled cases, study loopy BP and EP, tune parameters, and measure the accuracy–calibration–computation trade-off. These experiments are intended to quantify the improvement from smoothing and its associated computational cost.

7. Initial KPIs

7.1 Representation KPIs

- Number of skill, performance, and team nodes generated for a fixed tournament/window.
- Number of factors and edges in the corresponding factor graph.
- Growth of graph size as the number of matches/players increases.

7.2 Inference KPIs

- Predictive log-loss for win-probability quality.
- Brier score for probabilistic accuracy and calibration.
- Number of loopy-BP iterations required to meet the convergence criterion.
- Wall-clock inference time for exact/controlled inference, loopy BP, and EP.

7.3 Learning KPIs

- Held-out log-evidence or predictive log-likelihood for evaluating learned hyperparameters.
- Sensitivity of posterior and predictive performance to skill-dynamics and variance hyperparameters.

7.4 Statistical Reporting

At later milestones, results will be reported over multiple random seeds and/or synthetic populations and data splits, with mean and standard deviation. Where a claim of improvement between methods is made, an appropriate paired statistical test will be used. At M1 these are KPI definitions only; no numerical KPI results are claimed.

8. Dataset(s)

8.1 Primary Benchmark: ATP Historical Match Data

The primary benchmark is the ATP historical match dataset originally collected by Jeff Sackmann and packaged for this research by Landfried. It contains approximately 447,000 singles and doubles matches

from 1915–2020 involving more than 19,000 players. Available fields include participating players, singles/doubles indicator, tournament name and start date, tournament round, and playing surface.

8.2 Why the Dataset Fits

The chronological ordering of matches directly supports the temporal skill-evolution component of the Bayesian network. Its scale also provides a realistic setting for evaluating graph growth, convergence, predictive quality, and computational cost.

8.3 Synthetic Validation Dataset

A small synthetic dataset with a known skill trajectory will be generated in a later milestone. The purpose is to validate whether the implemented inference methods recover a controlled ground-truth trajectory before conclusions are drawn from real tennis data. Synthetic data is therefore a planned validation artefact, not an M1 result.

8.4 Data Sources

Primary ATP package: https://github.com/glandfried/tennis_atp/releases/download/atp/history.csv.zip

Original ATP source: https://github.com/JeffSackmann/tennis_atp

Official replication package/base-paper page: <https://www.jstatsoft.org/article/view/v112i06>

9. Experimental Plan

9.1 Planned Sequence

- Reproduce the 2025 base paper's controlled two-player, two-event example (Figure 3) by hand and in code to validate the factor-graph construction.
- Implement exact sum-product inference on controlled/tree-structured cases and validate it using the synthetic dataset with known ground truth.
- Implement loopy belief propagation and expectation propagation; first evaluate convergence and approximation behaviour on small controlled cases.
- Learn/tune hyperparameters using evidence or predictive likelihood, then run the full comparison among constant skill, Elo, forward-only TrueSkill, and TTT on the ATP benchmark.
- Collect the R/I/L KPIs and evaluate prediction quality, calibration, convergence, and computation time across multiple runs.
- Run robustness experiments by reducing data and adding controlled noise or other relevant perturbations; use an appropriate statistical significance analysis for method comparisons.
- Use the resulting evidence to refine the final methodological comparison with BONG. BONG will remain a reference point unless a later milestone explicitly defines and implements a fair experimental comparison.

9.2 Expected Experimental Logic

The experiments are designed to isolate where improvement comes from. Controlled synthetic cases test correctness against known behaviour; small cyclic cases test approximation and convergence; the ATP benchmark tests scalability and predictive usefulness; robustness experiments test whether conclusions persist under changed data quality or quantity. The final analysis will explicitly relate accuracy/calibration gains to inference cost rather than reporting accuracy alone.

10. Team Responsibilities

10.1 Work Allocation

The team will use a three-way workstream split so that every member owns a technically meaningful part of the project while cross-checking the complete R/I/L pipeline.

10.2 Shared Responsibility

All members are responsible for understanding the complete Representation–Inference–Learning pipeline and for cross-checking the consistency of assumptions, equations, implementation plans, KPI definitions, report, and video.

11. Timeline

11.1 Milestone Schedule

Milestone	Target date	Planned focus
M1 — Proposal & Scoping	15 September 2026	Problem, R/I/L mapping, baseline, SOTA/related work, datasets, KPIs, experiment plan, team roles.
M2 — Technical Foundation	11 October 2026	Bayesian network/factor graph, exact inference, Elo and forward-only TrueSkill baselines, synthetic generator, initial experiments.
M3 — Results & Analysis	1 November 2026	Synthetic + ATP experiments, inference comparison, robustness, statistical analysis, revised SOTA positioning.
M4 — Final Submission	22 November 2026	Complete theoretical analysis, experiments, KPI tables/plots, limitations, final SOTA positioning, reproducibility package.

11.2 M1 Status

M1 is a scoping milestone. The project defines the problem, graphical representation, inference methods, learning strategy, baselines, SOTA/related work positioning, datasets, KPIs, experimental plan, responsibilities, and timeline. Implementation, learned parameter values, and numerical experimental results are intentionally deferred to later milestones.

12. GitHub Repository

https://github.com/veera-bharakhada/g3_th_apl_gaming-rating_team

13. References

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